

Beyond Stable Answers: Auditing Explanation Faithfulness in a Construction-Safety Vision-Language Pipeline

Abstract

Vision-language models (VLMs) can support construction-safety monitoring by connecting natural-language safety rules to visual evidence. However, answer accuracy alone does not establish whether the evidence underlying a safety judgment is faithful or robust. This feasibility study adapts a six-dimensional explainable artificial intelligence evaluation framework, originally developed for tabular intrusion detection, to a VLM-grounded construction-safety pipeline. We evaluated Florence-2-base-ft on 163 held-out images from ConstructionSite 10k representing compliant scenes, personal protective equipment violations, fall hazards, and struck-by risks. Because Florence-2 does not provide native free-form visual question answering in this implementation, open-vocabulary grounding identified workers and rule-relevant safety objects, while a deterministic person-relative rule produced the compliance judgment. Explanation quality was tested through rule-aware region ranking, safety-object masking, severity-controlled perturbations, targeted and random-location occlusion, size-invariant centroid drift, bounding-box disappearance, worker-loss correction, bootstrap confidence intervals, and paired Wilcoxon tests. Area ranking selected the worker rather than the safety object in 50.3% of usable cases; rule-aware ranking reduced this to 0.6%. The corrected rule-aware top-region ablation changed 37.4% of judgments. Targeted object occlusion changed 39.2% of judgments and produced greater explanation drift than the strongest random-location control (0.198 versus 0.142; paired $n = 144$, $p = .0011$). Although intersection over union initially classified 25.0% of stable-answer cases as drifted, only 13.7% moved more than 0.2 of the image diagonal, revealing substantial small-box bias; 5.1% of targeted explanations disappeared entirely. These findings show that trustworthy construction-safety AI requires explanation-level evaluation, semantic region selection, size-aware metrics, and explicit failure accounting in addition to answer-level performance. The pipeline remains a research testbed requiring human oversight, not an autonomous safety system.